



EMMD: Question Bank on Unit-2

Q1. The probability distribution of X is given by

X	0	1	2	3	4
$P(X = x)$	k	$2k$	$3k$	$2k$	k

Find k , $P(X < 3)$, $P(1 \leq X \leq 4)$, $E(X)$, and $V(X)$.

Answer: $k = \frac{1}{9}$, $P(X < 3) = \frac{2}{3}$, $P(1 \leq X \leq 4) = \frac{8}{9}$, $E(X) = 2$, and $V(X) = \frac{4}{3}$.

Q2. A random variable X has the following probability distribution:

X	1	2	3	4	5	6	7
$P(X = x)$	k	$2k$	$3k$	k^2	$k^2 + k$	$2k^2$	$4k^2$

Find:

- (i) the value of k ,
- (ii) $P(X > 5)$,
- (iii) $P(1 \leq X \leq 5)$.

Answer: (i) $k = \frac{1}{8}$; (ii) $P(X > 5) = \frac{3}{32}$; (iii) $P(1 \leq X \leq 5) = \frac{29}{32}$.

Q3. For a continuous random variable X , the probability density function is

$$f(x) = \begin{cases} kx(2-x), & 0 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

Find:

- (i) the value of k ,
- (ii) mean,
- (iii) standard deviation.

Answer: (i) $k = \frac{3}{4}$; (ii) $E(X) = 1$; (iii) $\sigma = \frac{1}{\sqrt{5}} \approx 0.4472$.

Q4. The probability density function of a distribution is given as

$$f(x) = \lambda x e^{-x}, \quad x > 0.$$

Find:

- (i) the value of λ ,
- (ii) $P(2 < X < 5)$.

Answer: (i) $\lambda = 1$; (ii) $P(2 < X < 5) = 3e^{-2} - 6e^{-5} \approx 0.3656$.

Q5. If 20% of the bolts produced by a machine are defective, determine the probability that, out of 8 bolts chosen at random:

- (i) exactly 3 bolts are defective,
- (ii) no bolt is defective,
- (iii) at most 2 bolts are defective,
- (iv) at least 6 bolts are defective.

Answer: (i) 0.14680064; (ii) 0.16777216; (iii) 0.79691776; (iv) 0.00123136.

Q6. A pair of dice is thrown 20 times. If getting a doublet is considered a success, find the probability of

- (i) exactly 4 successes,
- (ii) at most 18 successes,
- (iii) no success.

Answer: (i) $\binom{20}{4} \left(\frac{1}{6}\right)^4 \left(\frac{5}{6}\right)^{16} \approx 0.202204$; (ii) using the complementary event, $P(X \leq 18) = 1 - P(X = 19) - P(X = 20) = 1 - \binom{20}{19} \left(\frac{1}{6}\right)^{19} \left(\frac{5}{6}\right) - \left(\frac{1}{6}\right)^{20} \approx 0.9999999999999972$; (iii) $\left(\frac{5}{6}\right)^{20} \approx 0.0260841$.

Q7. Find the mode of a binomial distribution whose mean and variance are 18 and 9, respectively.

Answer: $n = 36$, $p = \frac{1}{2}$, and mode = 18.

Q8. The average number of phone calls coming to a call centre is 5 per hour. Find the probability that during a particular hour there will be at most 2 phone calls.

Answer: $P(X \leq 2) = e^{-5} \left(1 + 5 + \frac{25}{2}\right) \approx 0.124652$.

Q9. If the probability that an individual suffers a bad reaction from a certain injection is 0.001, use the Poisson approximation to find the probability that, out of 2000 people:

- (i) exactly 3 will suffer a bad reaction,
- (ii) more than 2 will suffer a bad reaction.

Answer: With $\lambda = np = 2$: (i) $P(X = 3) = \frac{4}{3}e^{-2} \approx 0.180447$; (ii) $P(X > 2) = 1 - 5e^{-2} \approx 0.323324$.

Q10. Fit a Poisson distribution to the following data and calculate the theoretical frequencies:

x	0	1	2	3	4
f	122	60	15	2	1

Answer: $\lambda = 0.5$. The theoretical frequencies for $x = 0, 1, 2, 3, 4$ are approximately 121.306, 60.653, 15.163, 2.527, and 0.316, respectively (rounded: 121, 61, 15, 3, 0).

Q11. Fit a Poisson distribution to the following data and calculate the theoretical frequencies:

x	0	1	2	3	4
f	109	65	22	3	1

Answer: $\lambda = 0.61$. The theoretical frequencies for $x = 0, 1, 2, 3, 4$ are approximately 108.670, 66.289, 20.218, 4.111, and 0.627, respectively (rounded: 109, 66, 20, 4, 1).

Q12. The average number of misprints per page is 1.2. Find the probability that a page contains:

- (i) no misprint,
- (ii) exactly 2 misprints,
- (iii) at least one misprint.

Answer: (i) $e^{-1.2} \approx 0.301194$; (ii) $\frac{(1.2)^2}{2} e^{-1.2} \approx 0.216860$; (iii) $1 - e^{-1.2} \approx 0.698806$.

Q13. A light bulb manufacturing factory finds that 3 out of every 60 light bulbs are defective. Find the probability that the first defective light bulb will be found when the fifth bulb is tested. Also find the mean, variance, and standard deviation of the distribution.

Answer: $P(X = 5) = (0.95)^4(0.05) \approx 0.0407253$; mean = 20; variance = 380; standard deviation = $\sqrt{380} \approx 19.4936$.

Q14. A basketball player makes a basket with probability 0.7 on each attempt. Find the probability that the first successful basket occurs on the fourth attempt. Also find the mean, variance, and standard deviation. Explain why the mode of a geometric distribution is always 1.

Answer: $P(X = 4) = (0.3)^3(0.7) = 0.0189$; mean = $\frac{10}{7} \approx 1.4286$; variance = $\frac{30}{49} \approx 0.6122$; standard deviation = $\frac{\sqrt{30}}{7} \approx 0.7825$. The mode is 1 because $P(X = x) = (1 - p)^{x-1}p$ decreases as x increases.

Q15. An MNC conducted an aptitude test for 1000 candidates. The average score is 45 and the standard deviation is 25. Assuming a normal distribution, find:

- (i) the number of candidates whose score exceeds 60,
- (ii) the number of candidates whose score lies between 30 and 60.

Given:

$$A(0.6) = 0.2257.$$

Answer: (i) Approximately 274 candidates; (ii) approximately 451 candidates.

Q16. In a sample of 1000 cases, the mean of a certain test is 14 and the standard deviation is 2.5. Assuming a normal distribution, find:

- (i) how many students scored between 12 and 15,
- (ii) how many students scored less than 8 marks.

Given:

$$A(0.4) = 0.1554; \quad A(0.8) = 0.2881; \quad A(2.4) = 0.4918.$$

Answer: (i) Approximately 444 students; (ii) approximately 8 students.

Q17. Weights of 4000 students are normally distributed with mean 50 kg and standard deviation 5 kg. Find the number of students with weights:

- (i) less than 45 kg,
- (ii) between 45 kg and 60 kg.

Given:

$$A(1) = 0.3413; \quad A(2) = 0.4772.$$

Answer: (i) Approximately 635 students; (ii) approximately 3274 students.

Q18. In a certain company, 2000 LED bulbs are installed on each floor. If the LED bulbs have an average lifetime of 1000 burning hours with standard deviation 200 hours, find the number of LED bulbs that might be expected to fail within 700 hours. Assume a normal distribution.

Given:

$$A(1.5) = 0.4332.$$

Answer: Approximately 134 LED bulbs.